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Roll No. ..................

**6th Sem. / Civil Engg.**

**Subject : Steel Structure Design and Drawing**

Time : 6 Hrs. M.M. : 120

## SECTION-A

**Note:** Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 As per ISI rolled Steel beams sections are classified into.

a) two sections b) three sections

c) four sections d) five sections

Q.2 Met areas for plates connected by chain rivetting is

a) (b-nd) x d b) (b-nd) x t

c) (b-d) x t d) None of these

Q.3 When member is placed above the other and they are connected by means of rivets, the joint is known as

a) Lap joint

b) Double cover butt joint

c) Single cover butt joint

d) Both (a) and (b)

Q.4 The IS code which deals with steel structure is :

a) Is ! 456 b) Is ! 18 P3

c) Is ! 800 d) Is ! 2000

Q.5 Which of the following is not a type of column base.

a) Slab base b) Rolled base

c) Gusseted base d) All of the above

Q.6 Spacing of Purlins depend upon.

a) Type of roof covering

b) Slope of the truss

c) Both (a) and (b) d) None of these

## SECTION-B

**Note:** Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Define moment of resistance.

Q.8 ISSQ stand for __________.

Q.9 Write the formula for strength of fillet weld.

Q.10 Number of rivet required = Load in the member

Q.11 What are Purlins.

Q.12 The ratio of the rise to the span is called ________ of the roof truss.

## SECTION-C

**Note:** Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 List the assumptions in the theory of simple bending.

Q.14 Calculate the valve of 20mm diameter rivet in a lap joint connecting two plates 10mm and 15mm thick. Take $\tau vf=100\mu/mm^2$ and $\sigma pt=270\mu/mm^2$

Q.15 State various part of a roof truss.

Q.16 Explain the advantage and disadvantage of welded joint.

Q.17 Calculate the tensile strength of a tension member ISA 100x65x8mm, when it is connected by longer leg to a gusset plate using 20mm diameter rivets. Permissible tensile stress is 150 $\mu/mm^2$.

Q.18 What do you mean by fabrication.

Q.19 Write the procedure for desing of column.

Q.20 What are the various sections used in tension member.

Q.21 What are laterally restrained beams.

Q.22 What is the economic spacing of roof truss.

## SECTION-D

**Note:** Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Design a suitable section for a steel column of 3.50m unsupported length carrying a load of 500 KN. The column is continous throught more than two storeys and is restrained in position and direction at both ends.

Q.24 An ISMB 500 @ 852.5N/m carries a well of 22000 $\mu$/m (excluding its own weight) The beam is simply supported over an effective span of 6m calculate maximum bending stress and a unique shear stress.

Q.25 Calculate the gate compressive load carried by a double angle discontinous strut composed of 2 ISA 90x60x8 mm placed back to back and connected on both side of gusset plate 8mm thick by two rivets. The actual length of strut is 2.53m.

## PART-B

**Note:** Long answer type questions. Attempt any three questions out of four questions. (3x20=60)

Q.26 Draw the heel joint of a root struss with root drainage arrangements.

Q.27 Draw the front and side elevation of two unequal columns splicing arrangements with following data.

Lower column = ISHB 300@618.1μ/m

Upper column=ISHB 250@500.3μ/m

Cover Plates = 400mm x 250mmx20mm

Distribution plate = 300mmx250mmx20mm

Thickness of packaging plates=25mm

Cleat angles = ISA 75x75x8mm

Nominal diameter of rivet = 20mm

Q.28 Draw to a suitable scale the sectional plan front elevation and cross section of a sample plate girden from the following given data:

Clear Span = 12.0m

Web plate = 1000x12mm thick.

Top and bottom Hange cover plate=300mmx12mm thick

Flange angle=4-ISA 90x90x10mm.

Bearing plate = 200x300x20mm

Concrete block=300x300x200mm

Diameter of rivets=20mm.

Q.29 Draw a suitable scale front elevation and side elevation of a tunnel beam to beam connection. The main beam has two radianting beams connected to its web. The top of all the beams is at same level.

Main beam=ISMB 600@1202.7μ/m

Secondary beams=ISLB 300 @369.8μ/m

Cleat angles = 90x90x10mm

Nominal diameter of rivets = 20mm.